PROBLEM STATEMENT #17 | Healthcare & Telemedicine

Vaccination Cohort & Dose Scheduling System

1. Problem Context & Overview

A public health administration platform to organize demographic vaccination rollouts, enforce minimum interval days between doses, schedule center slots, and generate verifiable QR digital vaccination certificates.

Target Stakeholders / Actors: *Citizen Registrant, Vaccination Officer*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]
Description: The system shall enforce dose interval rules (e.g., minimum 28 days after Dose 1) before unlocking Dose 2 booking for a citizen.
Sample Acceptance Criteria: Pass: Dose 2 booking blocked if interval < 28 days; Fail: Early vaccination slot confirmed.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]
Description: The vaccination certificate verification endpoint shall authenticate digitally signed QR codes offline/online in under 150 ms.
Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**
  - >> Exactly 5 Functional Requirements (FR-001 to FR-005)
  - >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with
    ID, Type, Description, Priority, Acceptance Criteria, and Rationale.
- **UML Use-Case Diagram:**
  - >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.
- **Use-Case Flow Specification:**
  - >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
  - >> One Alternate Flow.